

IA générative

BONUS : Retour d'expériences

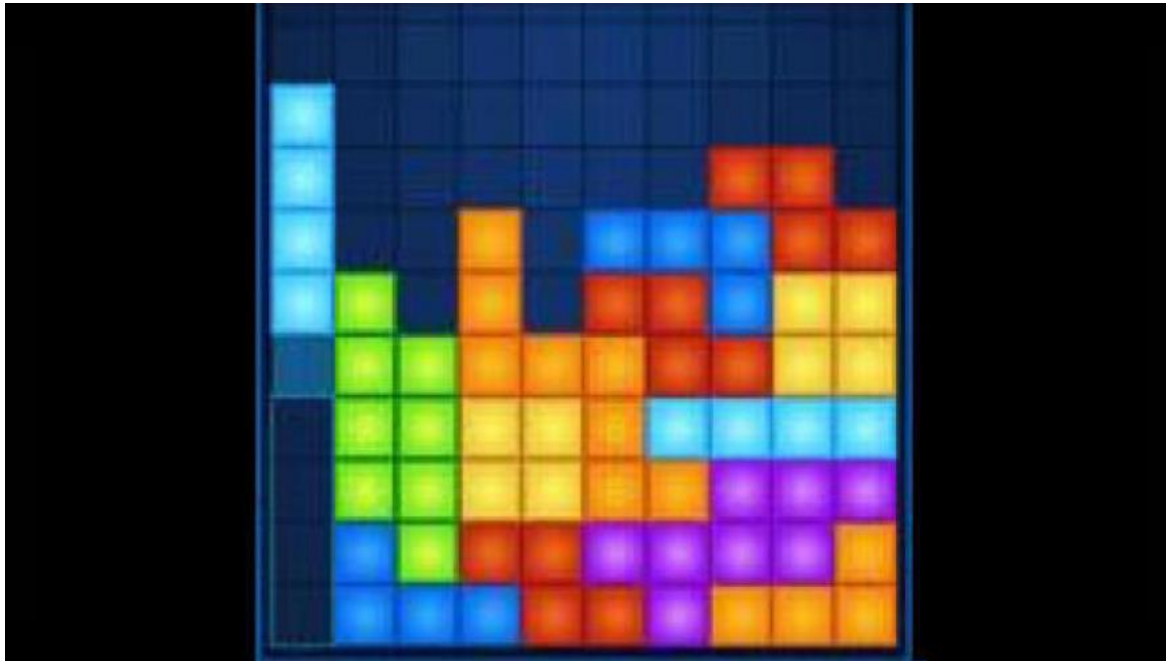


Présenté par **Morgan Gautherot**



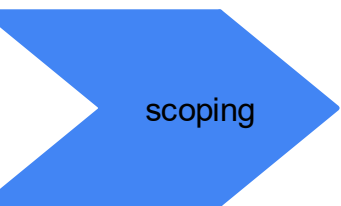
Exemple

Utiliser l'IA pour battre le champion du monde





Scoping



scoping

Objectif:

- Obtenir plus de 8 millions de points
- Détruire plus de 2 000 lignes



La vie d'un projet ML

scoping

Inventaire
des données

Pas de données



La vie d'un projet ML



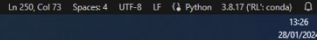
Pas de données



La vie d'un projet ML



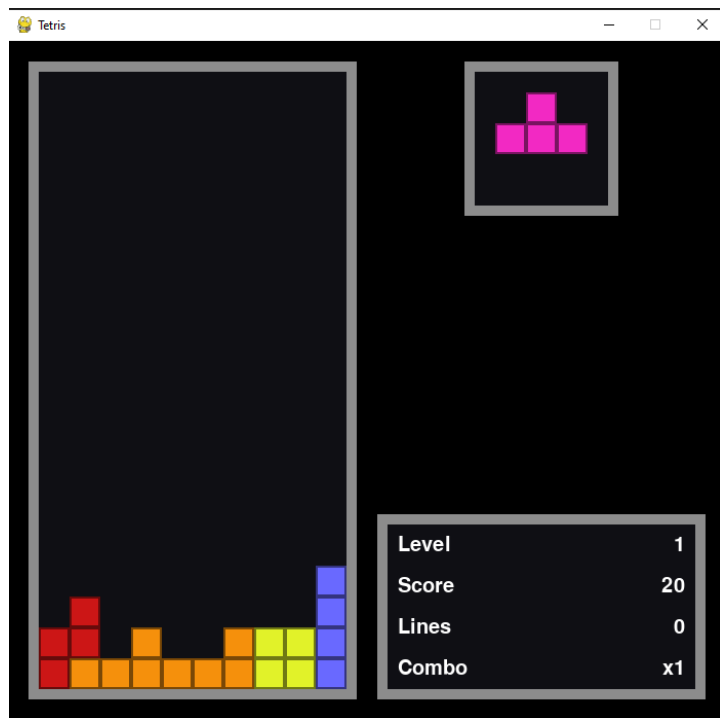
Création d'un système expert



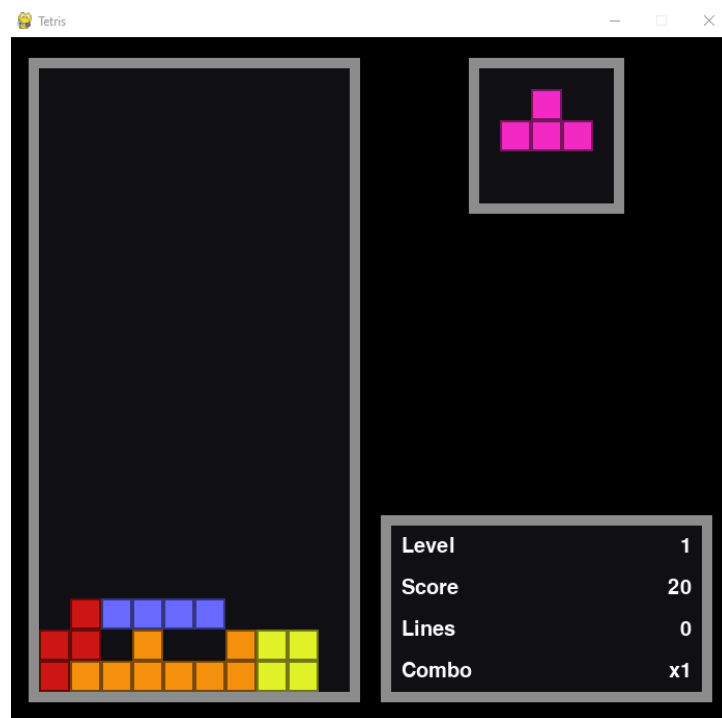


Créer une règle

Bien placé

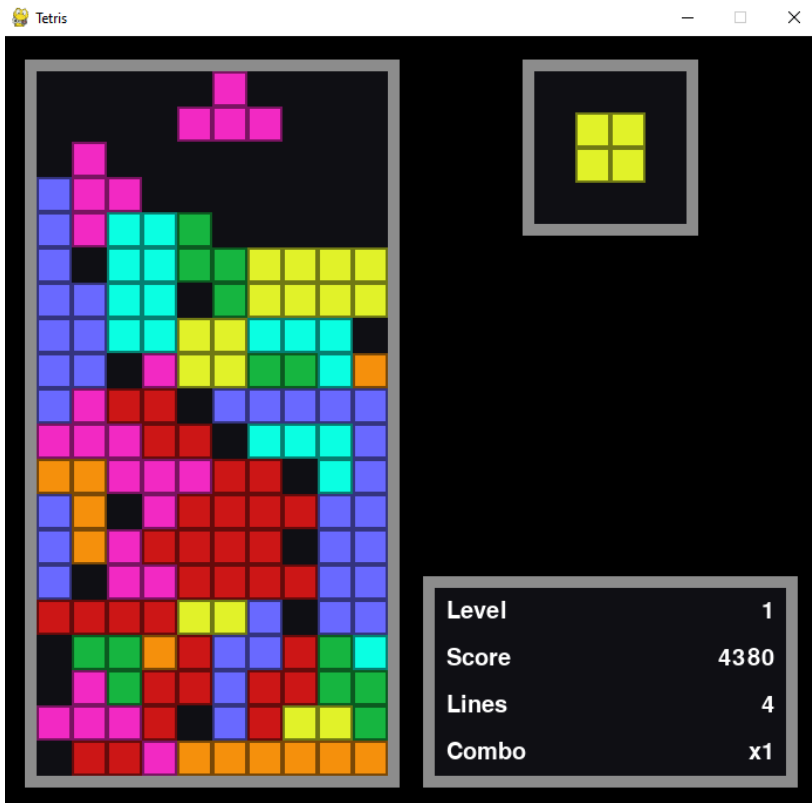


Mal placé





Ne pas laisser de trous



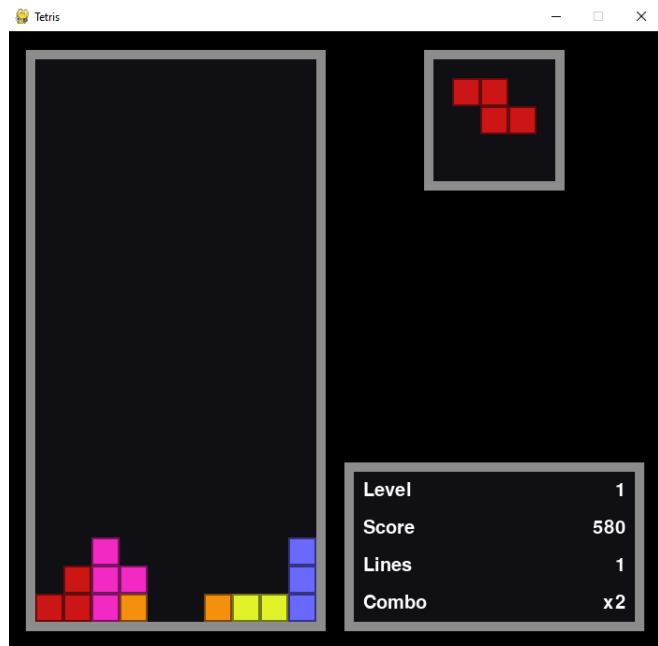


Fonction de coût

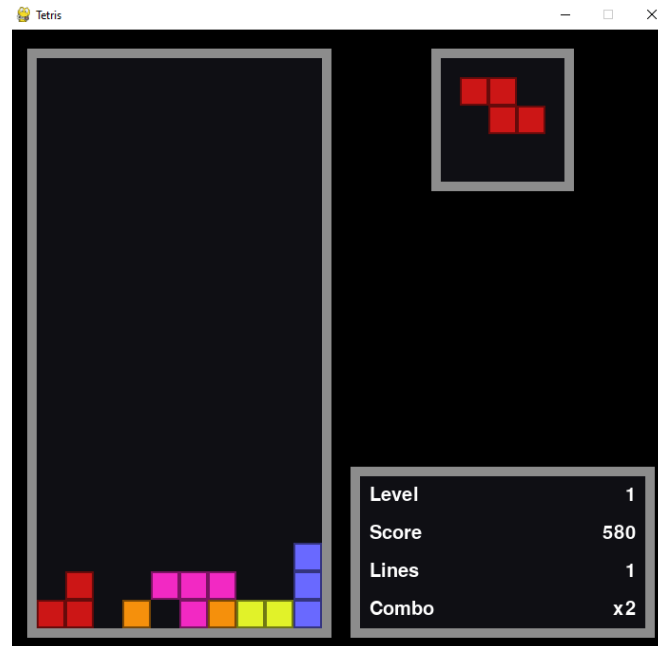
- Erreur = nombre de trous



Calculer l'erreur



Erreur = 0



Erreur = 1



Baseline en action

```
9
10
11 tetris = Tetris(take_picture = False,
12               training_id=3,
13               display=True,
14               seed=1132)
15
16 #bot = RandomBot(tetris)
17 bot = ExpertBot(tetris)
18
19 #bot = DeepBot(tetris,
20               # display=True)
21
22 while not(tetris.game_over):
23
24     # Time of the game
25     tetris.tetromino_falls_over_time()
26     matrix_and_tetromino = tetris.add_tetromino_to_game_board_matrix(tetris.current_tetromino,
27                                                                    tetris.game_board_matrix)
28     tetris.tetris_window.redraw(matrix_and_tetromino,
29                               tetris.next_tetromino)
30     #random_bot(tetris)
31     tetris.user_action()
32
33 bot.play()
34 #print(count_hole_number(tetris.matrix))
35 #custom_metric(tetris.matrix)
36 #system_expert(tetris)
37
38
39 print(tetris.tetris_score.score)
40 print(tetris.tetris_score.lines)
41
42
43 while True:
44     a = 1
45
46
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(myenv) PS C:\Users\Vaatherot\Documents\code\code_videos\Tetris_godmod\Tetris_godmod> python .\main_bot.py
pygame 2.5.2 (SDL 2.28.3, Python 3.11.6)
Hello from the pygame community. https://www.pygame.org/contribute.html
Traceback (most recent call last):
  File "C:\Users\Vaatherot\Documents\code\code_videos\Tetris_godmod\Tetris_godmod\main_bot.py", line 33, in <module>
    bot.play()
  File "C:\Users\Vaatherot\Documents\code\code_videos\Tetris_godmod\Tetris_godmod\src\bot.py", line 265, in play
    time.sleep(0.5)
KeyboardInterrupt
(myenv) PS C:\Users\Vaatherot\Documents\code\code_videos\Tetris_godmod\Tetris_godmod>
```

Ln 44, Col 10 Spaces: 4 UTF-8 CRLF Python 3.11.7 (RL\conda) 1344 28/01/2024

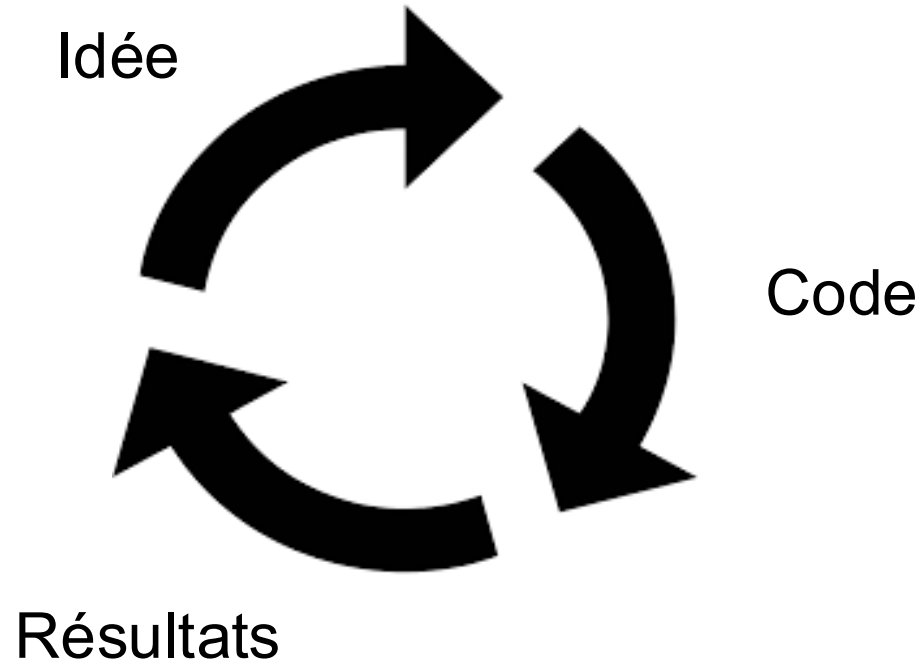


La vie d'un projet ML



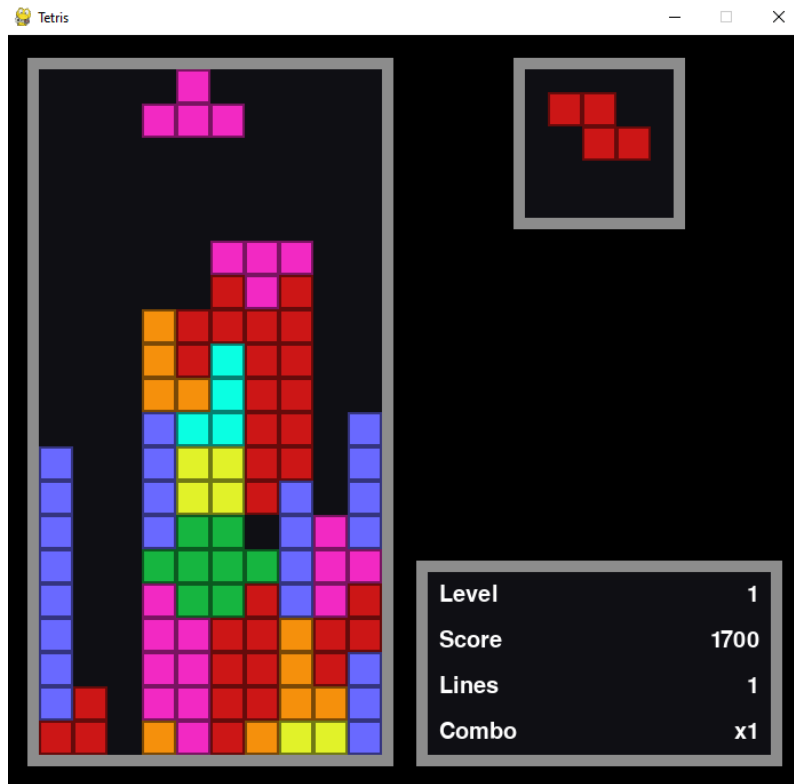


Itérations





Idée 1





Fonction de coût

- Erreur = nb_trous
+ hauteur_max
– hauteur_min



Résultats

```
def custom_metric(self, game_matrix:dict)->float:
    """
    Mix different cost function
    """
    nb_hole = self.count_hole_number(game_matrix)
    max_height = self.count_max_height(game_matrix)
    min_height = self.count_min_height(game_matrix)
    line = self.is_line(game_matrix)
    hole_colmun = self.count_hole_column(game_matrix)

    return nb_hole + max_height - min_height

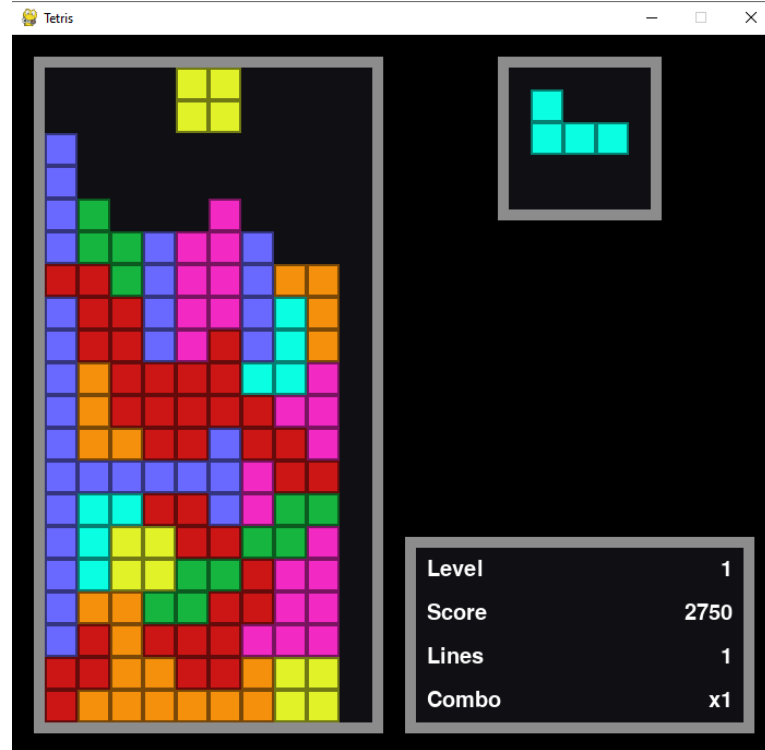
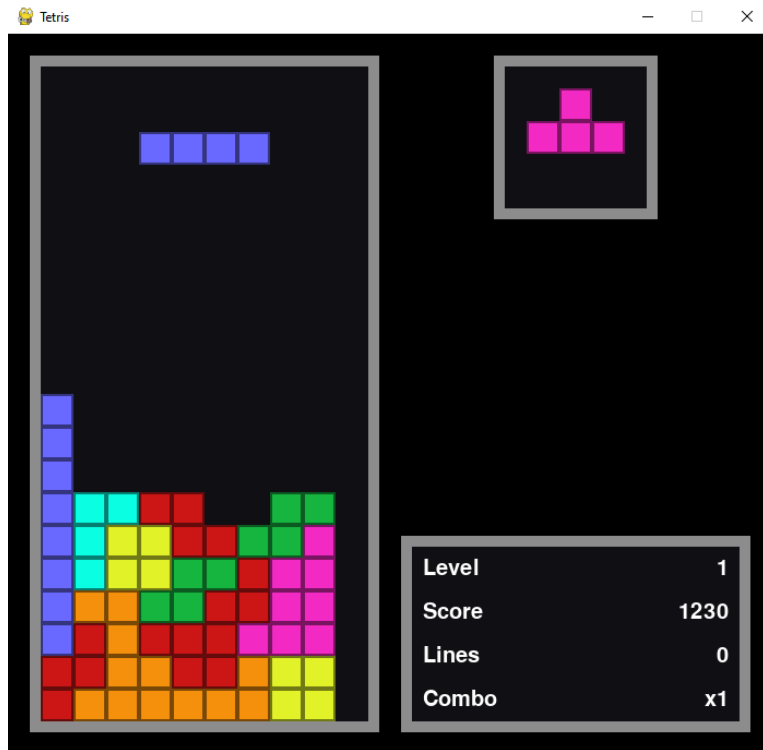
def count_hole_number(self, game_matrix:dict)->float:
    """
    Count the number of hole in the tetris matrix.
    A hole is an empty space cover by a tetrominoes
    """
    count_hole = 0
    number_hole = 0
    for width in range(config.MATRIX_WIDTH):
        number_hole = 0
        for height in range(config.MATRIX_HEIGHT-1, -1, -1):
            if game_matrix[height, width] is None:
                number_hole += 1
            if not(game_matrix[height, width] is None):
                count_hole += number_hole
```

File "C:\Users\Gautherot\Documents\code_videos\Tetris_godmod\tetris_godmod\src\window.py", line 85, in draw_surface
tetromino_block = self.block(board_game[(y, x)])
.....
File "C:\Users\Gautherot\Documents\code_videos\Tetris_godmod\tetris_godmod\src\window.py", line 118, in block
list(
File "C:\Users\Gautherot\Documents\code_videos\Tetris_godmod\tetris_godmod\src\window.py", line 120, in lambda
lambda c: min(255, int(c * 1)).
.....
KeyboardInterrupt
(myenv) PS C:\Users\Gautherot\Documents\code_videos\Tetris_godmod\tetris_godmod

Ln 118, Col 49 Spaces: 4 UTF-8 LF Python 3.8.17 (RL) conda 1351 28/01/2024

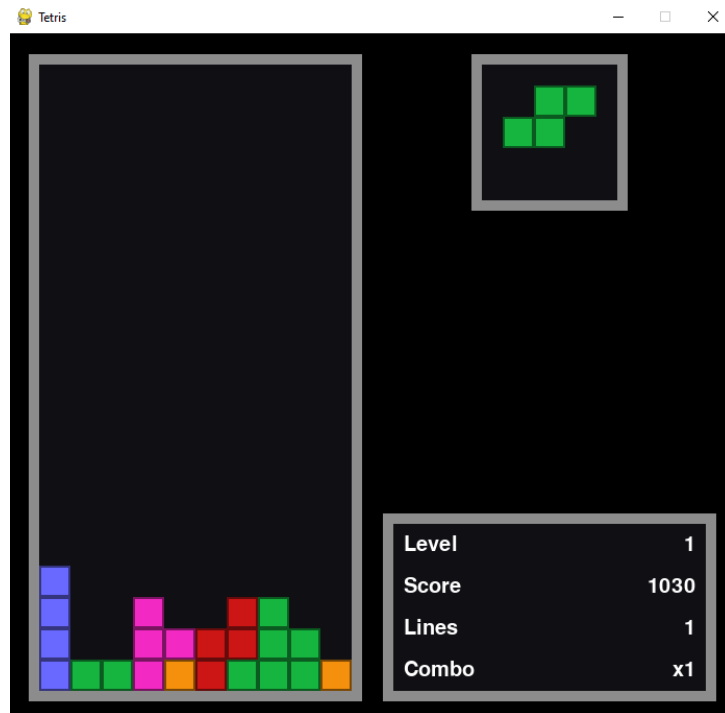
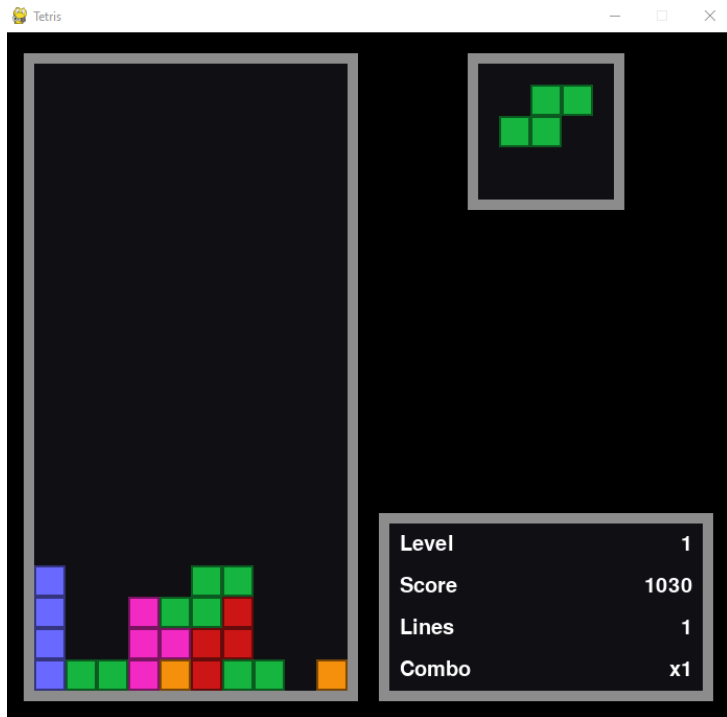


Idée 2



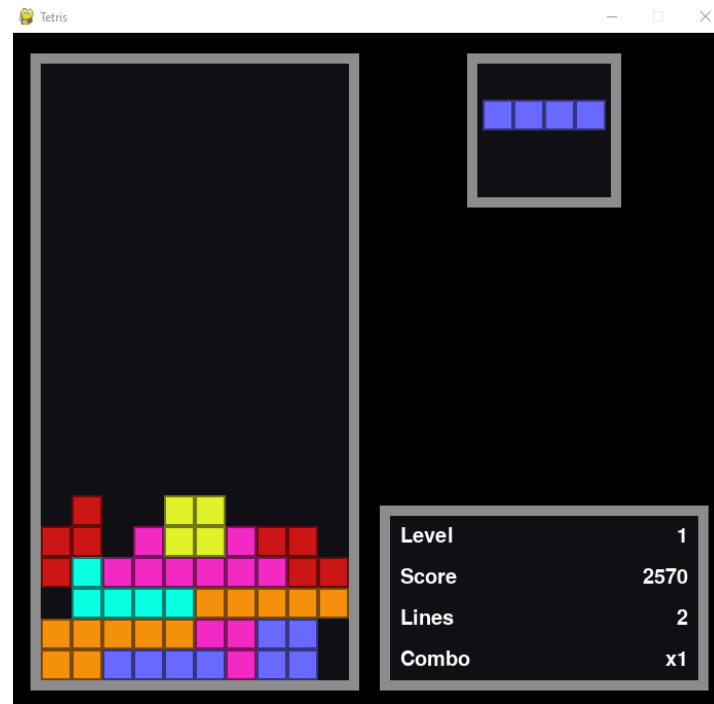
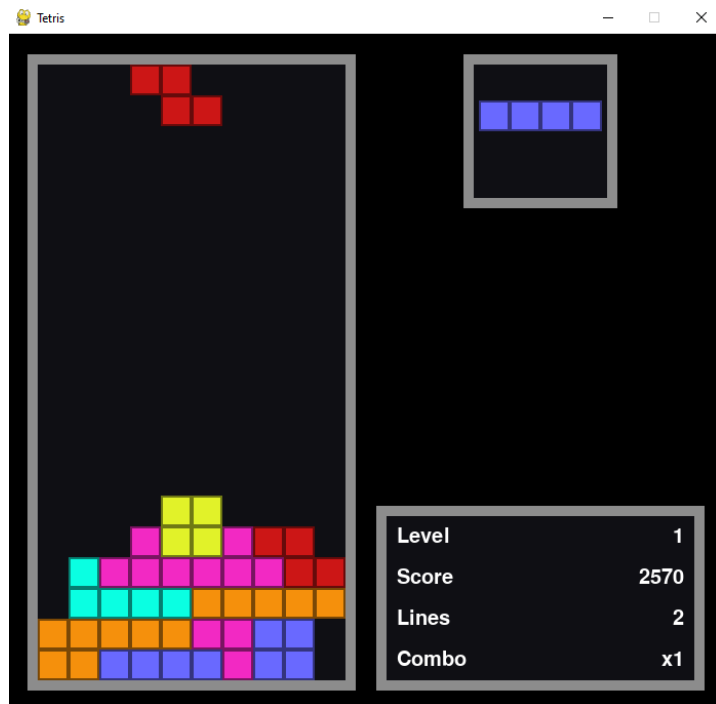


Idée 3



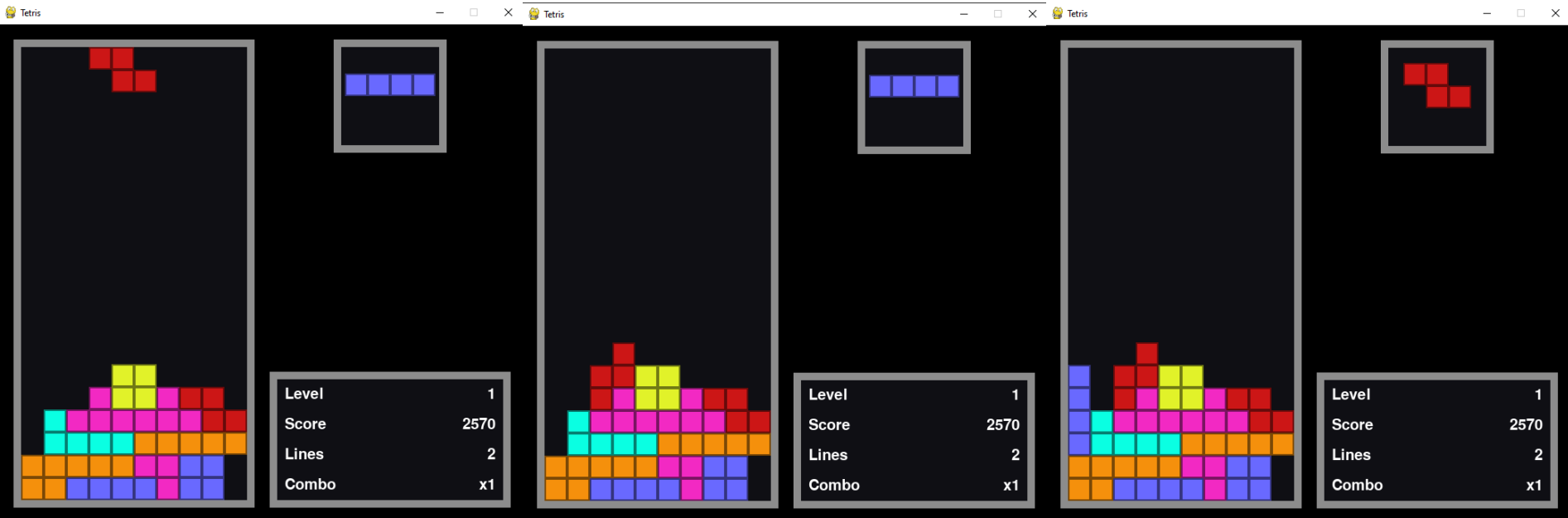


Idée 4





Idée 3





Double visualisation

```
rotation,
position_cost]],
columns=('id_column',
'id_rotation',
'cost'))

return move_cost_dataframe

def play(self)->None:
    """
    place the tetrominoes minimizing the hole
    """

    move_cost_dataframe = self.move_estimation(self.tetris.current_tetromino,
self.tetris.game_board_matrix,
recursivity_id=True)

    minimum_move_cost_dataframe = move_cost_dataframe[move_cost_dataframe['cost']==move_cost_dataframe['cost'].min()]
    #print(minimum_move_cost_dataframe)
    next_position = minimum_move_cost_dataframe.sample(1)
    #print(next_position)

    new_shape = self.tetris.current_tetromino.rotate(self.tetris.current_tetromino.tetromino_shape,
times=next_position.loc[0, 'id_rotation'])
    self.tetris.current_tetromino.tetromino_shape = new_shape

    self.tetris.current_tetromino.tetromino_position = (0, next_position.loc[0, 'id_column'])

    self.tetris.hard_drop()

    time.sleep(0.5)

# place the tetrominoes

(mysmv) PS C:\Users\Gautherot\Documents\code\code_videos\Tetris_godmod\tetris_godmod> python .\main_bot.py
pygame 2.5.2 (SDL 2.28.3, Python 3.11.6)
Hello from the pygame community. https://www.pygame.org/contribute.html
2570
2
Traceback (most recent call last):
  File "C:\Users\Gautherot\Documents\code\code_videos\Tetris_godmod\tetris_godmod\main_bot.py", line 43, in <module>
    while True:
KeyboardInterrupt
(mysmv) PS C:\Users\Gautherot\Documents\code\code_videos\Tetris_godmod\tetris_godmod>
```



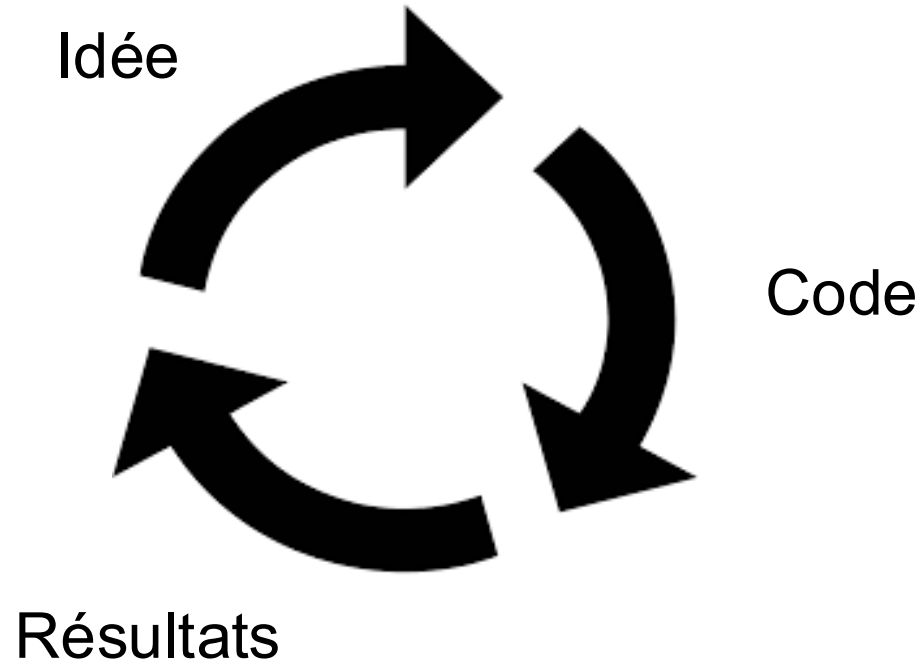
Résultats

Score: 1 447 207

Nombre de lignes : 812

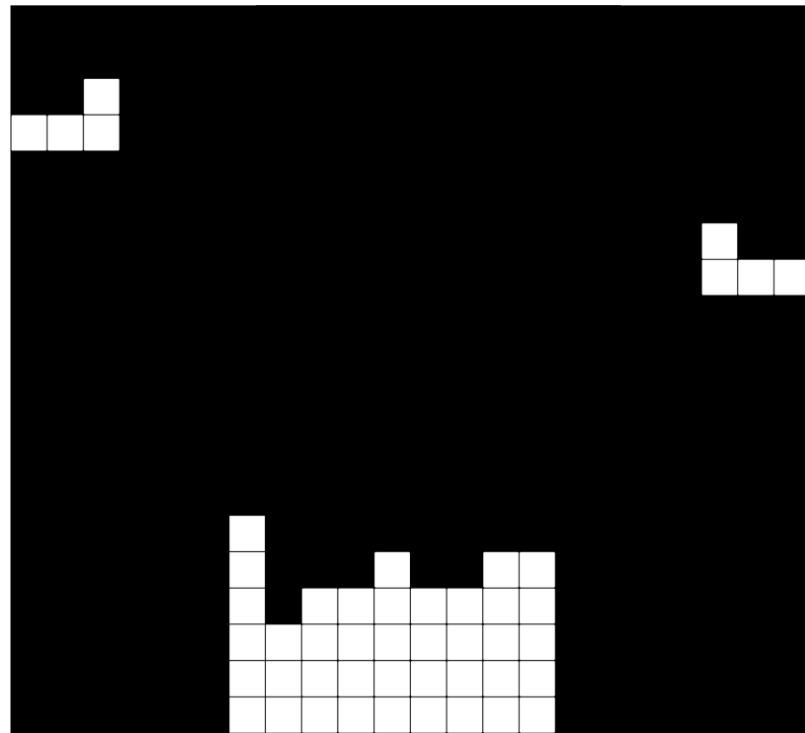
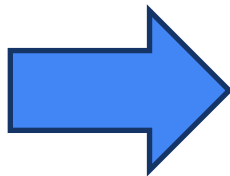
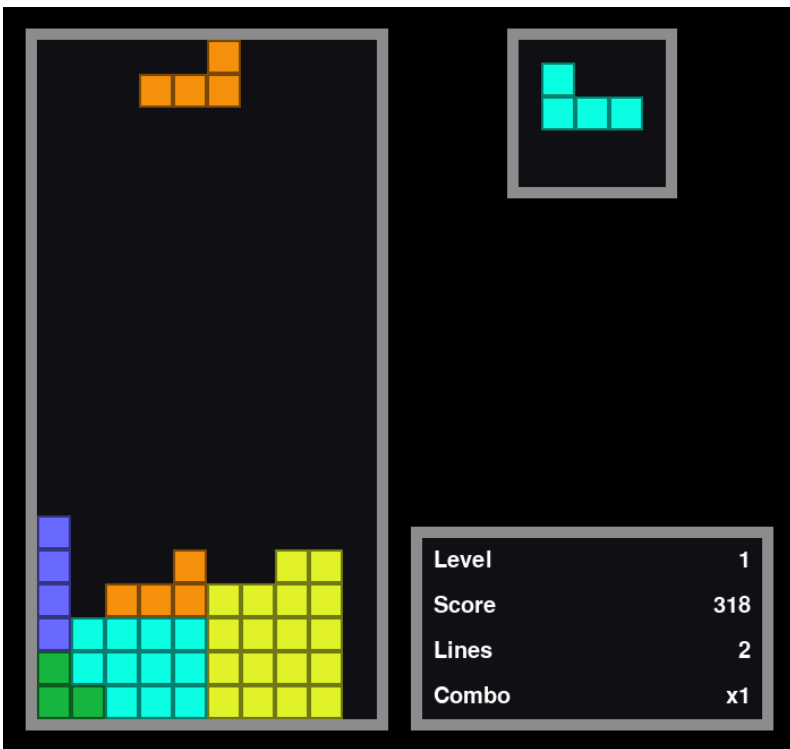


Itérations





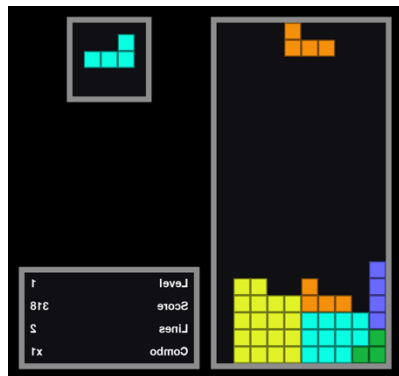
Enregistrement des données d'entrées



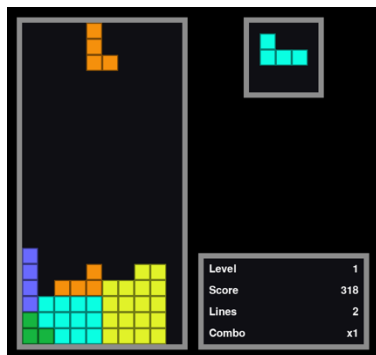


Enregistrement des données de sorties

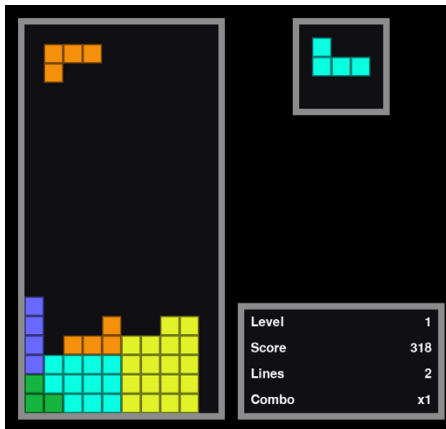
Etat initial



Rotation = 2



Colonne = 1



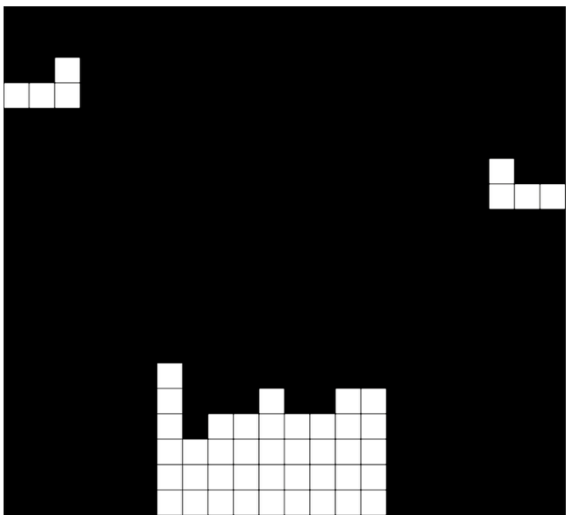
Etat final



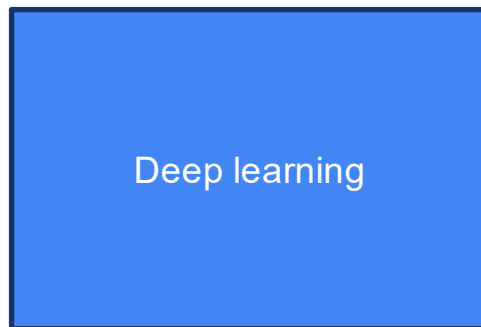


Entraînement d'un modèle

Données d'entrées



Modèle



Données de sortie

Rotation = 2

Colonne = 1



Quantité des données

Nombre images

Performances

Temps

5 000

Nulle

2 heures

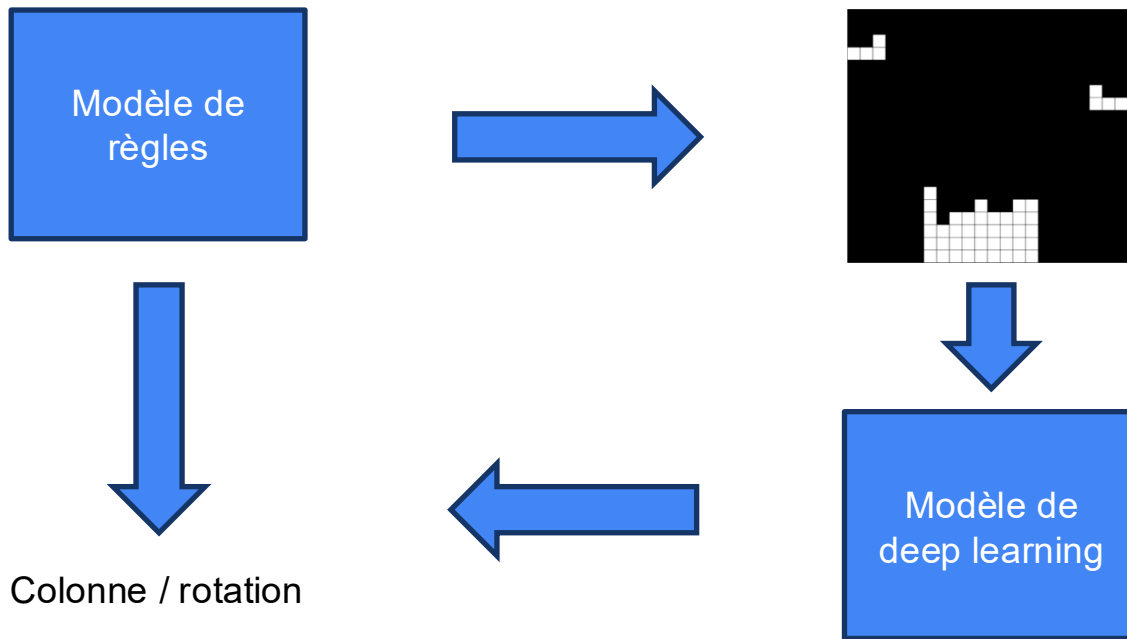
50 000

Très faible

2 jours



Génération de données





Quantité des données

Nombre images

Performances

Temps

10 000

Nulle

5 minutes

100 000

Très faible

1 heure

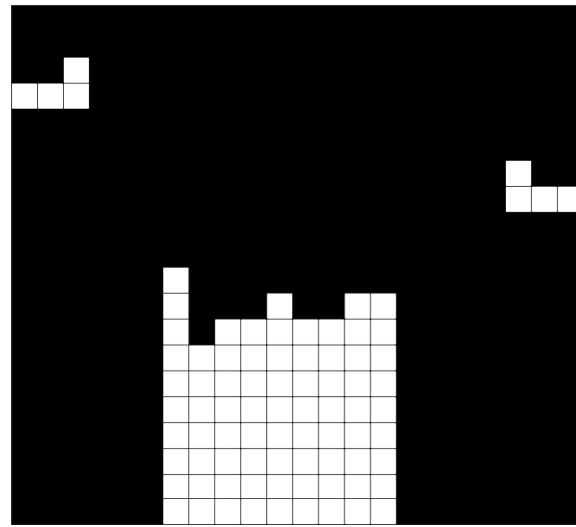
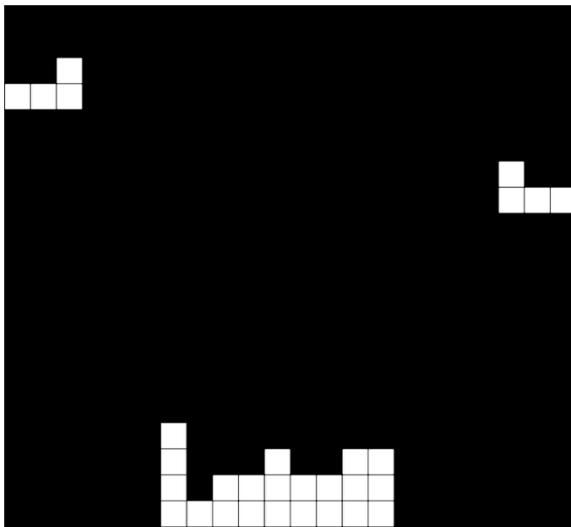
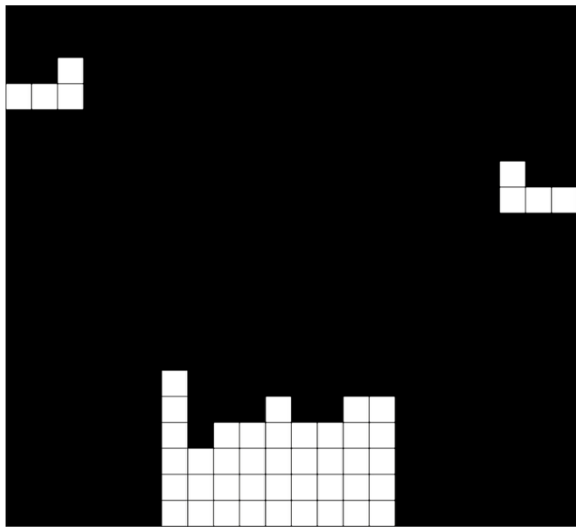
2 000 000

Faible

1 journée



Data augmentation





Quantité des données

	Nombre images	Performances	Temps
	10 000	Nulle	5 minutes
	100 000	Très faible	1 heure
	2 000 000	Faible	1 journée
Data augmentation	30 000 000	Moyenne	



Résultats

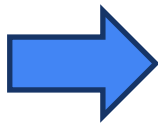
Score: 750 811

Nombre de lignes : 580

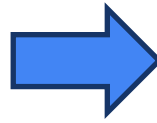


Qualité des données

Données



Modèle de
machine learning



Prédiction





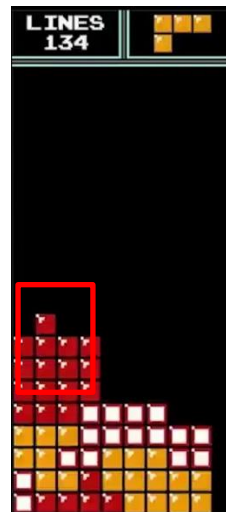
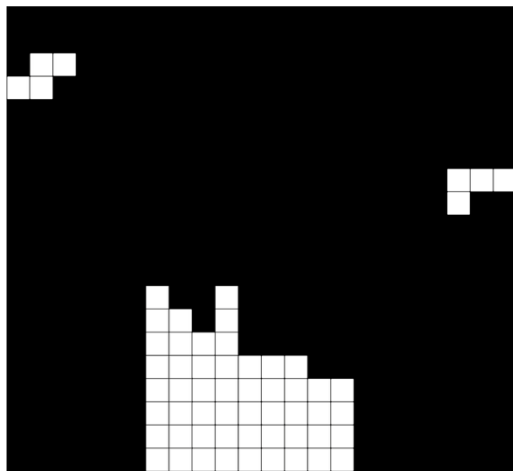
Parties de professionnelles





Création du dataset

Données d'entrée

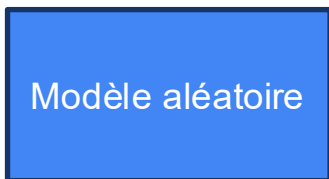


Rotation = 1

Colonne = 1

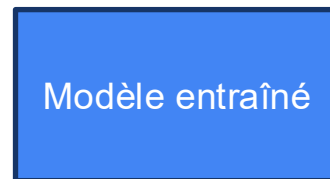


Fine tuning



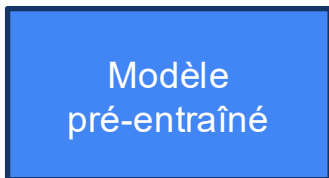
Performance nulle

Entraînement



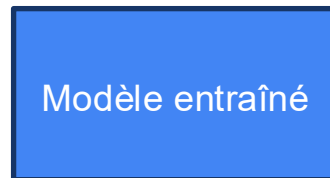
Performance moyenne

Fine tuning



Performance moyenne

Entraînement

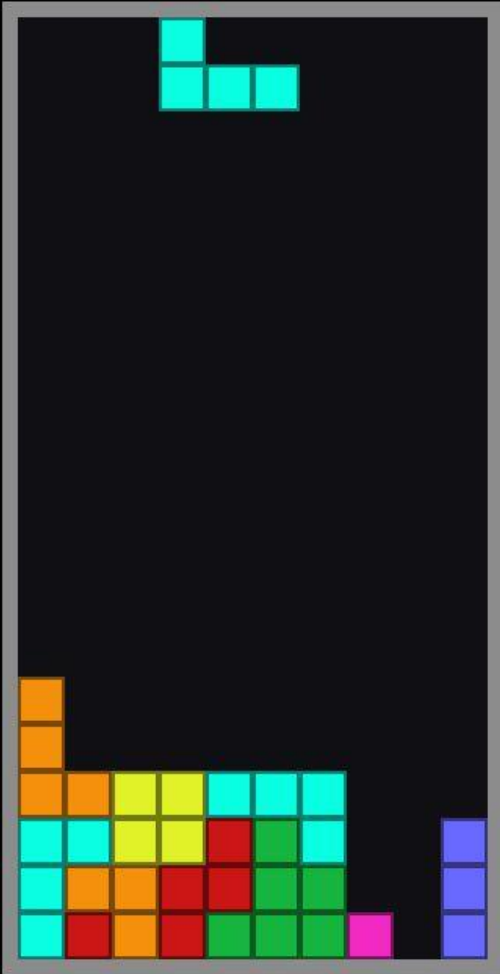


Performance incroyable



Résultats

Tetris



Level	202
Score	12926820
Lines	2011
Combo	x2



La vie d'un projet ML

